

CHP-5 Electric Energy

General Physics 2

2026-09-27

Energy of a System of Point Charges

Electric Energy

In the previous chapter, we discussed the potential energy of a point charge placed in the electric field of a given charge distribution.

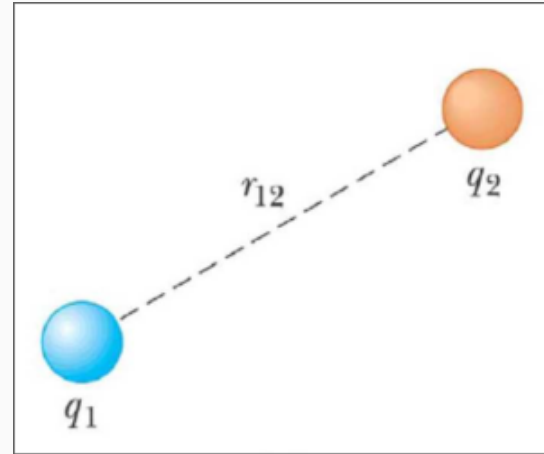
Now let us consider the potential energy of the charge distribution itself.

How much work is required to assemble a system of charges, starting with the charges very far apart?

Energy of Two Point Charges

Imagine two charges q_1 and q_2 separated by a distance r . Taking the potential energy to be zero at infinite separation,

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

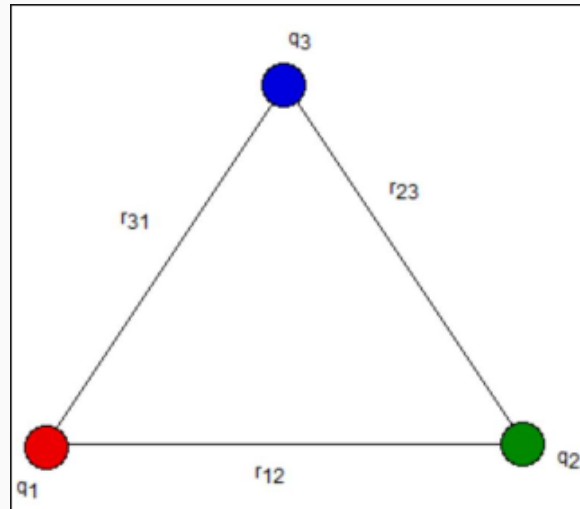


This is the external work required to bring q_1 slowly from infinity to a distance r from the fixed charge q_2 , or vice versa.

It is a **mutual potential energy**: it belongs to the pair of charges.

What Happens with More Charges?

Consider three charges q_1 , q_2 , and q_3 , with pairwise separations r_{12} , r_{13} , and r_{23} .



Add the interaction energy of each distinct pair:

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}}$$

Each pair is counted **once**. We can rewrite the same energy using the potential at each charge due to all the other charges.

$$\begin{aligned} U = & \frac{1}{2}q_1\left(\frac{1}{4\pi\epsilon_0}\frac{q_2}{r_{12}} + \frac{1}{4\pi\epsilon_0}\frac{q_3}{r_{13}}\right) \\ & + \frac{1}{2}q_2\left(\frac{1}{4\pi\epsilon_0}\frac{q_1}{r_{12}} + \frac{1}{4\pi\epsilon_0}\frac{q_3}{r_{23}}\right) \\ & + \frac{1}{2}q_3\left(\frac{1}{4\pi\epsilon_0}\frac{q_1}{r_{13}} + \frac{1}{4\pi\epsilon_0}\frac{q_2}{r_{23}}\right) \end{aligned}$$

Each parenthesis is the potential at that charge due to the **other charges**. The factor $\frac{1}{2}$ removes the double counting of each pair.

Energy of a System of Point Charges

Let $V_{\text{other}}(i)$ be the potential at charge q_i due to all the other charges. For three charges,

$$U = \frac{1}{2}q_1V_{\text{other}}(1) + \frac{1}{2}q_2V_{\text{other}}(2) + \frac{1}{2}q_3V_{\text{other}}(3)$$

Generalizing to any number of charges,

$$U = \frac{1}{2} \sum_i q_i V_{\text{other}}(i)$$

This gives the work required to bring the charges from very large separations to their final positions.

A Note About Self-Energy

The preceding formula gives the **interaction energy** between point charges.

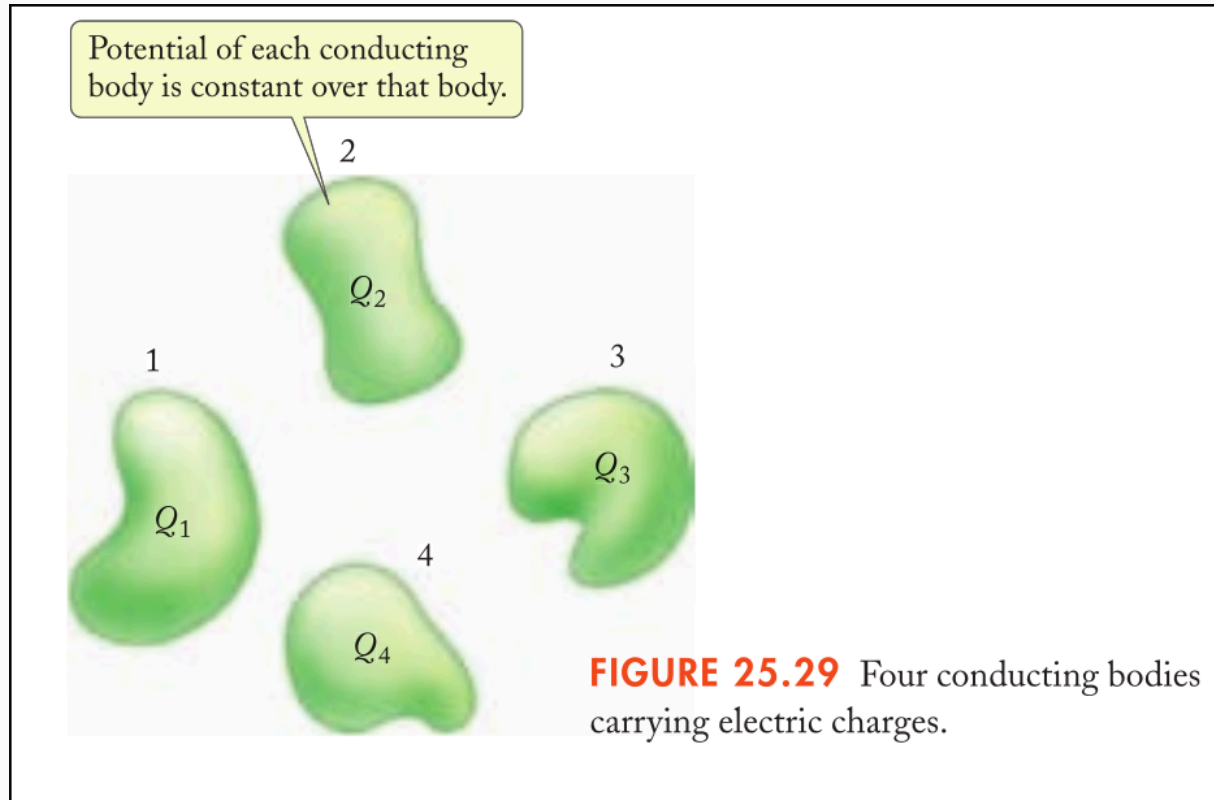
It does not include the work required to assemble each individual point charge from infinitesimal pieces of charge. That contribution is called **self-energy**.

In these point-charge calculations, we ignore the self-energies of the individual particles, such as electrons. Their treatment lies beyond the scope of this discussion.

Energy of a System of Conductors

From Point Charges to Conductors

Suppose several conductors carry charges $Q_1, Q_2, Q_3, \dots$ and have potentials $V_1, V_2, V_3, \dots$.



The potential at a charge element dQ_1 includes contributions from:

- the other charge elements on conductor 1;
- all the charge elements on the other conductors.

Its contribution to the total energy is

$$\frac{1}{2}V_1dQ_1$$

Here V_1 includes the potential due to conductor 1 as well as the other conductors.

Energy of the Conductors

In electrostatic equilibrium, each conductor is an equipotential. Therefore,

$$\frac{1}{2} \int V_i dQ_i = \frac{1}{2} V_i Q_i$$

Summing the contributions from all conductors gives

$$U = \frac{1}{2} Q_1 V_1 + \frac{1}{2} Q_2 V_2 + \frac{1}{2} Q_3 V_3 + \dots$$

Equivalently,

$$U = \frac{1}{2} \sum_i Q_i V_i$$

Example: A Charged Metallic Sphere

A metallic sphere of radius R carries a charge Q uniformly distributed over its surface. How much electric energy is stored in this configuration?

For a single isolated conductor,

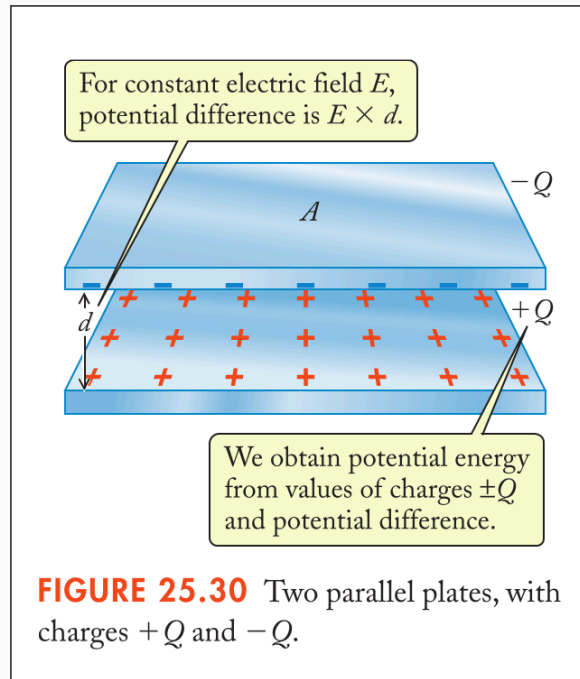
$$U = \frac{1}{2}QV, \quad V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

Hence,

$$U = \frac{1}{2}QV = \frac{Q^2}{8\pi\epsilon_0 R}$$

Example: Two Parallel Metallic Plates

Two large parallel metallic plates, each of area A , are separated by a distance d . They carry charges $+Q$ and $-Q$.



- What is the electric energy of this distribution? Neglect edge effects.

Let $Q_1 = Q$ and $Q_2 = -Q$, with potentials V_1 and V_2 .

$$\begin{aligned} U &= \frac{1}{2}Q_1V_1 + \frac{1}{2}Q_2V_2 \\ &= \frac{1}{2}QV_1 - \frac{1}{2}QV_2 \\ &= \frac{1}{2}Q(V_1 - V_2) \end{aligned}$$

Potential Difference Between the Plates

Between the plates,

$$E = \frac{\sigma}{\epsilon_0}, \quad \sigma = \frac{Q}{A}$$

Moving from the positive plate to the negative plate is moving along the electric field, so

$$V_2 - V_1 = - \int_1^2 \vec{E} \cdot d\vec{l} = -Ed = -\frac{Qd}{\epsilon_0 A}$$

The stored energy is therefore

$$U = \frac{1}{2}Q(V_1 - V_2) = \frac{Q^2 d}{2\epsilon_0 A}$$

Energy in the Electric Field

Energy Density Between the Plates

Rewrite the energy using $E = \frac{Q}{\epsilon_0 A}$:

$$U = \frac{1}{2}\epsilon_0\left(\frac{Q}{\epsilon_0 A}\right)^2 Ad = \frac{1}{2}\epsilon_0 E^2 (Ad)$$

Since Ad is the volume between the plates, the energy per unit volume is

$$u = \frac{1}{2}\epsilon_0 E^2$$

This expression describes the electric energy as distributed through the space between the plates.

Energy in an Electric Field

Although we obtained this result for parallel plates, the electric-field energy density applies to a general field in vacuum:

$$u = \frac{1}{2}\epsilon_0 E^2$$

Integrate the energy density over the volume of interest:

$$U = \int \frac{1}{2}\epsilon_0 E^2 dV$$

Here dV denotes a volume element. For the total field energy, integrate over all space.

Is the Energy in the Charges or the Field?

We have described energy in terms of charges and potentials,

$$U = \frac{1}{2} \sum_i q_i V_{\text{other}}(i),$$

and in terms of the electric field,

$$U = \frac{1}{2} \epsilon_0 \int E^2 dV.$$

The field description is useful because fields can carry energy through regions with no local charge, as in radio waves. Electromagnetic waves also carry magnetic-field energy.

Comparing the Two Descriptions

Both approaches give the same electrostatic energy when self-energy is treated consistently.

For ideal point charges, the field integral includes divergent self-energy contributions. The point-charge pair formula excludes them.

We will compare the two approaches for a **uniformly charged solid sphere**, where the continuous charge distribution has a finite total energy.

First assemble the sphere from thin spherical layers. Then integrate the field energy inside and outside it.

Example: A Uniformly Charged
Solid Sphere

First Method: Assemble Spherical Layers

Consider a sphere of radius R with charge Q uniformly distributed throughout its volume. Its charge density is

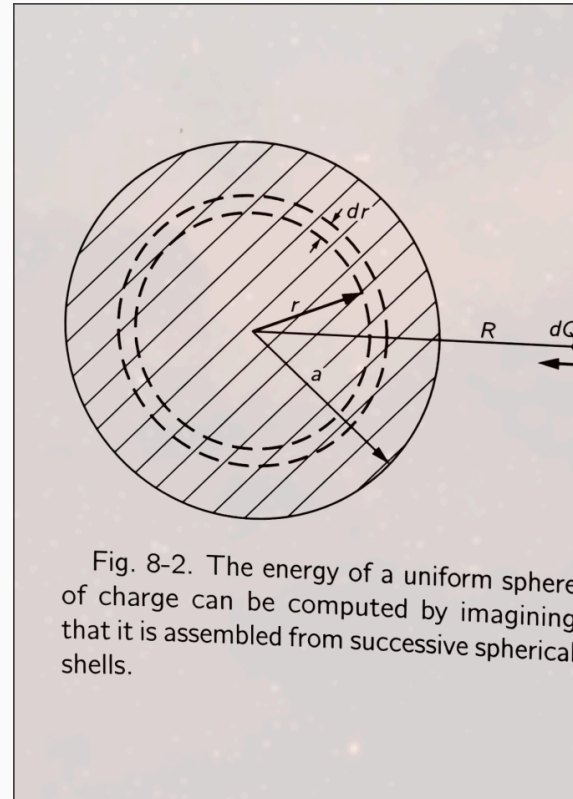
$$\rho = \frac{Q}{\frac{4\pi R^3}{3}} = \frac{3Q}{4\pi R^3}$$

Suppose the sphere has been built up to radius r . Its charge is

$$Q(r) = \rho \frac{4\pi r^3}{3}$$

The next thin spherical layer, of thickness dr , has charge

$$dQ = \rho 4\pi r^2 dr$$



Work Required to Add One Layer

The potential at radius r due to the already assembled sphere is

$$V(r) = \frac{Q(r)}{4\pi\epsilon_0 r}$$

Bringing the next infinitesimal layer from infinity requires work

$$dU = V(r)dQ = \frac{Q(r)dQ}{4\pi\epsilon_0 r}$$

Substituting the expressions for $Q(r)$ and dQ ,

$$dU = \frac{1}{4\pi\epsilon_0 r} \left(\rho \frac{4\pi r^3}{3} \right) (\rho 4\pi r^2 dr) = \frac{4\pi\rho^2}{3\epsilon_0} r^4 dr$$

Integrate from the Center to the Surface

To assemble the complete sphere, integrate from $r = 0$ to $r = R$:

$$U = \int_0^R \frac{4\pi\rho^2}{3\epsilon_0} r^4 dr = \frac{4\pi\rho^2 R^5}{15\epsilon_0}$$

Using $\rho = 3\frac{Q}{4\pi R^3}$,

$$U = \frac{4\pi}{15\epsilon_0} \left(\frac{3Q}{4\pi R^3} \right)^2 R^5$$

$$U = \frac{3}{5} \frac{Q^2}{4\pi\epsilon_0 R}$$

Second Method: Integrate the Field Energy

From Gauss's law, the radial field of the uniformly charged solid sphere is

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3} \quad \text{for } 0 \leq r \leq R,$$

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad \text{for } r \geq R.$$

The volume of a thin spherical shell is

$$dV = 4\pi r^2 dr$$

Compute the two contributions separately:

$$U = U_{\text{in}} + U_{\text{ext}}$$

Energy Outside the Sphere

$$\begin{aligned}U_{\text{ext}} &= \frac{1}{2}\epsilon_0 \int_R^\infty E(r)^2 4\pi r^2 dr \\&= \frac{1}{2}\epsilon_0 \int_R^\infty \left(\frac{Q}{4\pi\epsilon_0 r^2} \right)^2 4\pi r^2 dr \\&= \frac{Q^2}{8\pi\epsilon_0} \int_R^\infty \frac{1}{r^2} dr\end{aligned}$$

Since $\int_R^\infty r^{-2} dr = \frac{1}{R}$,

$$U_{\text{ext}} = \frac{Q^2}{8\pi\epsilon_0 R}$$

Energy Inside the Sphere

$$\begin{aligned}U_{\text{in}} &= \frac{1}{2}\epsilon_0 \int_0^R E(r)^2 4\pi r^2 dr \\&= \frac{1}{2}\epsilon_0 \int_0^R \left(\frac{Qr}{4\pi\epsilon_0 R^3} \right)^2 4\pi r^2 dr \\&= \frac{Q^2}{8\pi\epsilon_0 R^6} \int_0^R r^4 dr\end{aligned}$$

Since $\int_0^R r^4 dr = \frac{R^5}{5}$,

$$U_{\text{in}} = \frac{Q^2}{40\pi\epsilon_0 R}$$

Both Methods Give the Same Result

Add the exterior and interior contributions:

$$\begin{aligned}U &= U_{\text{ext}} + U_{\text{in}} \\&= \frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{R} + \frac{1}{5R} \right) \\&= \frac{3Q^2}{20\pi\epsilon_0 R}\end{aligned}$$

$$U = \frac{3}{5} \frac{Q^2}{4\pi\epsilon_0 R}$$

This is the same energy obtained by assembling the sphere from spherical layers.